TECHNICAL EVALUATION REPORT · AIML WITH PYTHON

Student Performance Analysis and Prediction System

A data exploration and Linear Regression study using the UCI Student Performance dataset.

Aditya Rathore
BCA, Artificial Intelligence and Data Science
Dev Bhoomi Uttarakhand University

Training Program & Date
September 2026 Batch
Submission Report, October 2026

649	4	0.86	0.75
student records	*model inputs*	*test R-squared*	*test MAE (marks)*

Educational analysis notice: Predictions are mathematical estimates that must be reviewed by a person. This system is not for high-stakes decisions or autonomous evaluation.

Made with GAMMA

Executive summary and objectives

Examines how study time, absences, and earlier grades relate to final grade G3 (0 to 20) with descriptive analysis and Linear Regression.

0.75

MAE marks

130 test records

1.17

RMSE marks

held-out test

0.86

Test R-squared

0.85

Mean five-fold CV R-squared

Results describe this dataset and split, not other settings.

Objectives

- Inspect data quality.
- Summarize study time, absences, and grades.
- Train on G1, G2, studytime, absences.
- Evaluate on held-out and five folds.
- Explain failure points and why human review remains necessary.

Boundary: the 10/20 review flag is a demonstration aid only, not a school policy, diagnosis, or decision about a student; not for high-stakes use.

Made with GAMMA

DATA ARCHITECTURE

Dataset, source, and preparation

UCI Student Performance dataset (Repository ID 320), containing 649 secondary education records from two schools in Portugal, fetched via `ucimlrepo`.

Variable	Role & Scale	Mean	Median	Std Dev	Range
studytime	Input (Category 1–4)	1.93	2.00	0.83	1 to 4
absences	Input (Count)	3.66	2.00	4.64	0 to 32
G1	Input (Grade 0–20)	11.40	11.00	2.75	0 to 19
G2	Input (Grade 0–20)	11.57	11.00	2.91	0 to 19
G3	Target (Grade 0–20)	11.91	12.00	3.23	0 to 19

Middle half of final grades (G3 interquartile range) spans 10 to 14 marks. Clean dataset: zero missing values and zero duplicate rows across 33 initial fields.

Partitioning Protocol

An 80/20 train-test partition was executed with `random_state=42`, yielding 519 training records and 130 held-out evaluation records. Because data integrity was verified with no missing entries, no imputation or record deletion was performed.

Structural Constraints

Earlier term marks G1 and G2 precede final mark G3 and serve as strong predictors from earlier grading periods. Studytime is recorded as an ordinal category (1: under 2 hours, 2: 2–5 hours, 3: 5–10 hours, 4: over 10 hours weekly), not as exact continuous hours.

Made with GAMMA

EXPLORATORY DATA ANALYSIS

Study time and final-grade distribution

Full dataset, before the split; descriptive only, not causal.

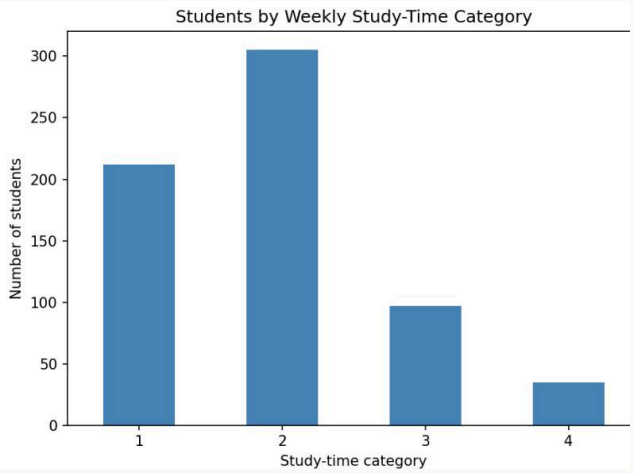


Figure 1. Number of records by study-time category. Category 2 is most common; counts are uneven. Category 1 is under 2 hours weekly, category 4 over 10.

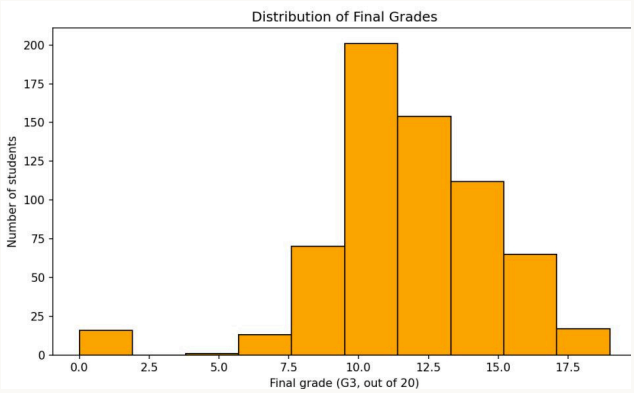


Figure 2. Distribution of G3. Values concentrate mid-scale, with records at zero and near the top; one average cannot describe all students.

Made with GAMMA

Absences and study-time comparisons

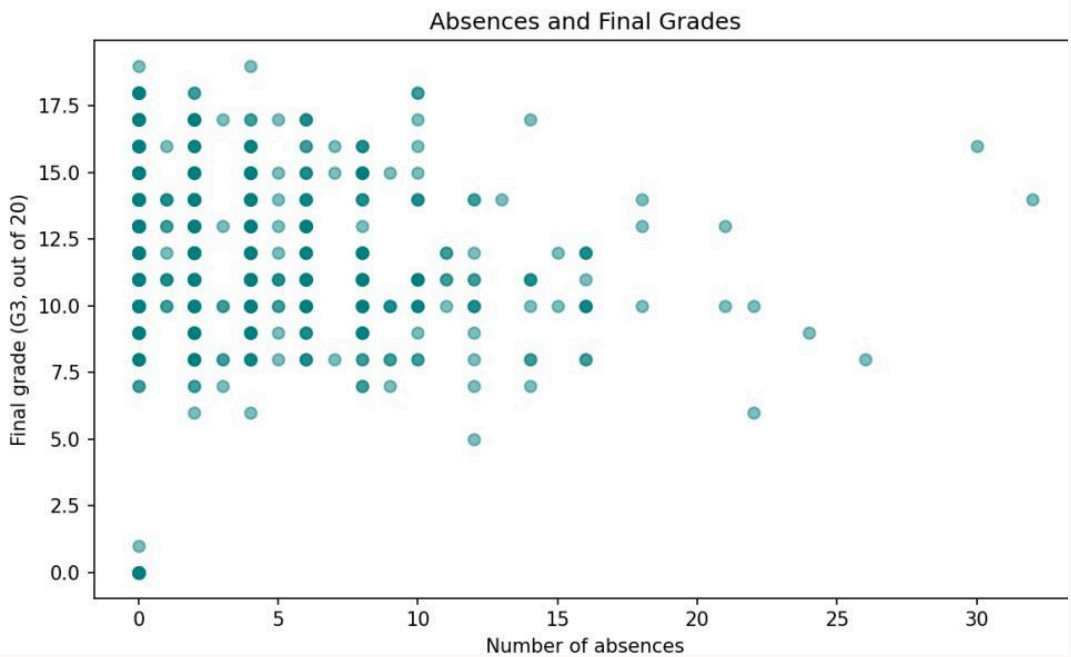


Figure 3. Absences and final grade. Points overlap because grades are discrete; wide variation at similar absence counts; few observations at high absences. Correlation of absences with G3 is -0.09.

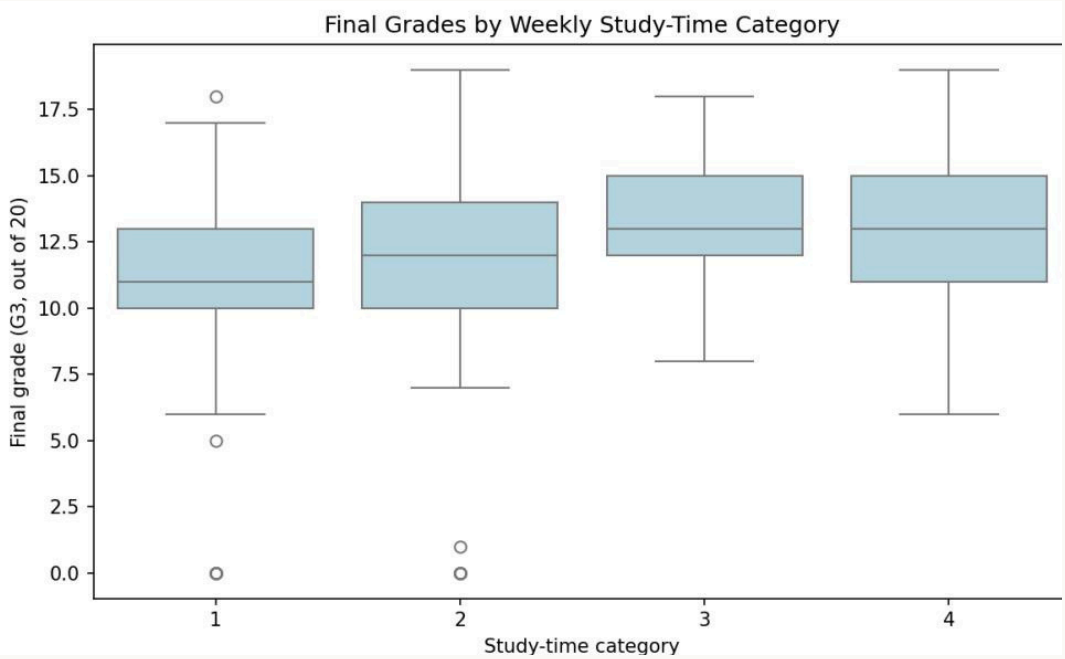


Figure 4. G3 by study-time category. Medians are higher for categories 3 and 4 than 1, but distributions overlap and group sizes differ. This does not show that study time alone causes a grade change.

Made with GAMMA

MODELING PROTOCOL

Correlation and methodology

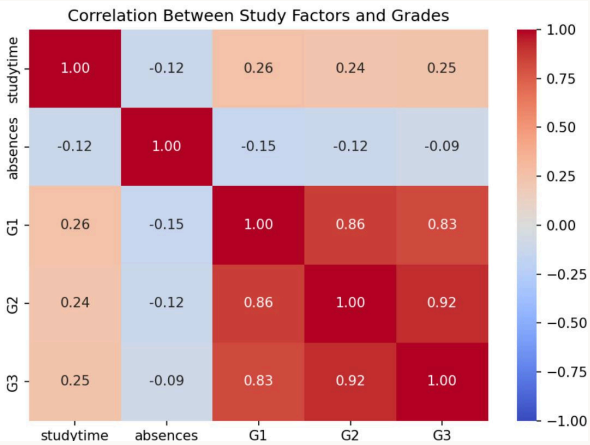


Figure 5. Correlation matrix (-1 to +1). Correlation is not causation.

Observed Pearson Correlations

Pairwise linear associations with final grade G3:

- **Second-period mark (G2):** 0.92 correlation
- **First-period mark (G1):** 0.83 correlation
- **Study time category:** 0.25 correlation
- **Recorded absences:** -0.09 correlation

G1 and G2 correlate strongly with each other at 0.86, introducing collinearity that complicates isolating individual regression coefficients.

Regression Specification

Linear Regression implemented via scikit-learn fits target G3 using four features: `studytime`, `absences`, `G1`, and `G2`. Five-fold cross-validation is evaluated strictly inside the 519-record training split to protect against leakage prior to held-out test scoring.

Notebook Inference Example

A sample record with `studytime=2`, `absences=3`, `G1=12`, and `G2=13` yields an estimated G3 of 13.3 / 20. This single point is an illustrative software verification output, not an aggregate performance claim.

Made with GAMMA

Evaluation

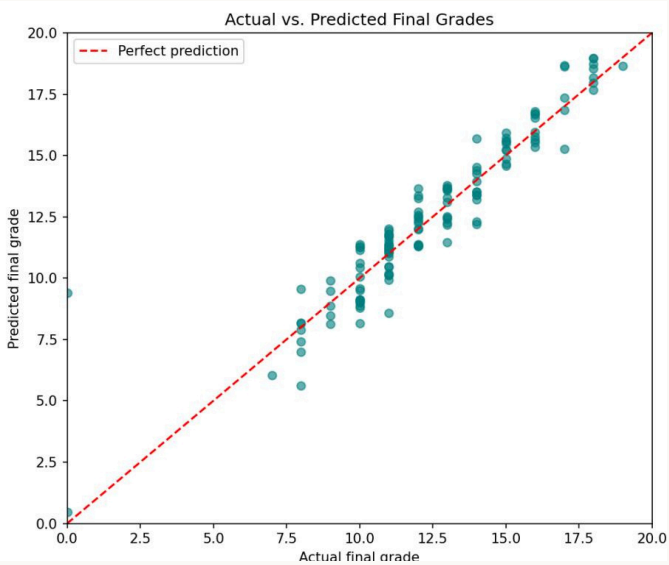


Figure 6. Actual versus predicted grades on held-out records; the red dashed line is perfect agreement. The model follows the broad pattern, with errors on individual records.

Cross-Validation Stability

Fold	R-squared
Fold 1	0.88
Fold 2	0.75
Fold 3	0.86
Fold 4	0.88
Fold 5	0.87
Mean CV R-squared	**0.85**

Held-Out Test Results

Held-out test (130 records): MAE 0.75 marks, RMSE 1.17 marks, R-squared 0.86.

Review Threshold Screening

A demonstration threshold of 10/20 flagged 27 test predictions for human review; this is not a school rule or diagnosis.

Made with GAMMA

SYNTHESIS & GOVERNANCE

Limitations, future work, conclusion, and references

Identified Limitations

- **Geographic scope:** Data reflects two regional secondary schools in Portugal; findings do not generalize across distinct curricula.
- **Measurement fidelity:** Study time is grouped into four broad bands; absences are raw counts without illness distinction.
- **Precursor grades:** G1 and G2 are strongly correlated (0.86), so their individual contributions are difficult to separate. The model assumes linear relationships, and evaluation on one random split may not capture performance at grade extremes.
- **Sample scale:** Test performance was evaluated on one held-out split of 130 records; additional validation on other samples is needed.

Proposed Future Work

Evaluation on multi-school external cohorts, testing regularized or tree-based non-linear estimators, residual analysis across lower grade tiers, continuous study-hour logging, and establishing review thresholds collaboratively with educators.

Summary Conclusion

Across the held-out partition, the system yielded MAE 0.75, RMSE 1.17, R^2 0.86, and mean CV R^2 0.85. The pipeline serves as an educational benchmark and analytical study; it is entirely unsuitable for autonomous or high-stakes student evaluations.

Reproducibility

Jupyter notebook, raw dataset export, dependency specifications, README documentation, and visual figures are maintained in the project's open repository.

References

Cortez, P. (2008). *Student Performance*. UCI Machine Learning Repository, dataset 320. doi:10.24432/C5TG7T

UCI Documentation: archive.ics.uci.edu/dataset/320

Codebase: github.com/AdiRathoreX/student-performance-analysis-prediction

Made with GAMMA